

Learning-Rate Schedules and Peak-LR Sensitivity in a Tiny Character-Level Transformer: A Small CPU Study

Pyrite-66R

Stemma seed corpus

<https://github.com/mihir-chauhan/research-harness-papers/tree/main/papers/learning-rate-schedules-on-a-character-level-trans>

Abstract—We compare four learning-rate treatments of one hand-written AdamW on a 2-layer, $d = 128$ character-level GPT (421632 parameters) trained for 1500 steps on Tiny Shakespeare: constant, linear warmup plus cosine decay, step decay, and a reimplemented Schedule-Free AdamW. Each is run at five peak learning rates (3×10^{-4} to 3×10^{-2}) with 3 seeds, and we define peak-LR sensitivity as the worst-minus-best validation loss over the grid. At the best of the three main peak rates, warmup+cosine (1.67 nats at 3×10^{-3}) beats constant LR (1.76 at 10^{-3} ; Welch $p = 0.001443$), but Schedule-Free AdamW is better still (1.63 at 10^{-2}). Our hypothesis that Schedule-Free is less sensitive to the peak LR than warmup+cosine is refuted: over the three main rates its sensitivity is 0.25 nats against 0.09 for cosine. Most of the large gaps between schedules at high peak rates are a warmup effect rather than a schedule effect: constant and step decay as specified have no warmup and reach 2.42 and 2.35 at 10^{-2} , while giving constant LR a 100-step warmup removes most of the gap (1.81). With three seeds, one model size and one budget, these are small-scale findings, and the schedule-free result depends on our own implementation and untuned warmup.

Index Terms—learning-rate schedules, warmup, cosine decay, schedule-free optimisation, hyperparameter sensitivity, character-level language model

I. Introduction

The learning-rate schedule is one of the cheapest knobs in neural-network training and one of the easiest to get wrong. Transformers were introduced with an explicit warmup-then-decay rule [1]; linear warmup had earlier been recommended for large-minibatch SGD [2]; and cosine annealing [3] combined with decoupled weight decay in AdamW [4], [5] is today’s default recipe. Why warmup helps is still debated: one explanation attributes it to the high variance of Adam’s adaptive step early in training [6], others analyse how it lets a network tolerate a larger peak learning rate [7] or how much of it GPT training can do without [8]. A recent alternative removes the decay schedule altogether: Schedule-Free AdamW [9] replaces

it by iterate averaging and is advertised as needing no stopping-time-dependent schedule.

Optimiser benchmarks repeatedly find that conclusions depend on the tuning protocol and schedule [10], [11]. A practitioner’s question that these large studies leave open is more modest: if I train a small model for a short, fixed budget and cannot afford to tune the peak learning rate carefully, which schedule loses least when the peak rate is wrong? We answer it for a deliberately small, fully reproducible setting that runs on a laptop CPU: a two-layer character-level GPT on Tiny Shakespeare for 1500 steps.

Contributions.: (i) A controlled comparison of four schedules, all sharing one hand-written AdamW, over five peak learning rates and 3 seeds each, with a defined sensitivity metric. (ii) Evidence that, in this setting, much of the apparent advantage of cosine decay over constant and step schedules at large peak rates is explained by warmup, which the baselines as originally specified lack; we separate the two with a warmup ablation. (iii) A negative result: Schedule-Free AdamW reaches the lowest loss but is not less sensitive to the peak LR than warmup+cosine.

Hypotheses: (registered before the runs in proposal.md). H1: at each system’s best main peak LR, warmup+cosine has lower validation loss than constant LR. H2: Schedule-Free AdamW has lower peak-LR sensitivity than warmup+cosine. H3: at the largest main peak LR (10^{-2}), removing warmup from cosine hurts.

II. Related Work

a) **Schedules and warmup.**: Cosine annealing was proposed together with warm restarts for SGD in [3]; we use its single-cycle form without restarts, as is common for language models. Warmup in the large-batch regime is motivated in [2], and for Adam by the variance of the adaptive learning rate in the first steps [6]. For GPT-style models, [7] study the mechanisms by which warmup permits larger peak learning rates, and [8] analyse when warmup can be reduced or removed. Our warmup ablation is a small-scale probe of the same phenomenon in the

Pyrite-66R is the codename of an autonomous study, not a person. Produced by the Stemma autonomous AI research pipeline; the topic was chosen to cover the field. Every reported result is traced to logged runs in the linked repository and every reference was checked against arXiv, Crossref, DataCite or OpenAlex.

specific setting of cross-schedule comparison, where the presence of warmup is a confound.

b) Decay phases.: The cooldown (decay) stage of warmup-stable-decay schedules has been analysed in [12]. Our step-decay and cosine schedules both contain a decay phase, but we do not study the dynamics of that phase; we only compare final loss.

c) Schedule-free optimisation.: Schedule-Free SGD/AdamW [9] interpolates between an averaged iterate and a base iterate and is evaluated at the average, removing the need to choose a decay horizon. We reimplement the AdamW variant from the algorithm description (not from the authors' code), including our choice of warmup; differences from the reference implementation may exist.

d) Benchmarks.: Large-scale benchmarks [10], [11] stress equal tuning budgets and show that optimiser rankings are sensitive to the protocol. Our study asks a complementary, narrower question: sensitivity to a single hyperparameter, the peak learning rate, at a tiny scale.

III. Method

All systems use the same optimiser and differ only in the learning-rate multiplier (or, for Schedule-Free, in the iterate used for evaluation). Let g_t be the clipped minibatch gradient at step $t = 1, \dots, T$ ($T = 1500$), η the peak learning rate, and γ_t the learning rate applied at step t .

a) AdamW (shared base).: With $\beta_1 = 0.9000$, $\beta_2 = 0.9500$, $\epsilon = 10^{-8}$:

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t, \quad v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2, \quad (1)$$

$$\theta_t = (1 - \gamma_t \lambda) \theta_{t-1} - \gamma_t \frac{m_t / (1 - \beta_1^t)}{\sqrt{v_t / (1 - \beta_2^t)} + \epsilon}, \quad (2)$$

with decoupled weight decay $\lambda = 0.1000$ applied to weight matrices only and global gradient-norm clipping at 1.

b) Constant.: $\gamma_t = \eta$ (no warmup, as the seed specifies).

c) Warmup+Cosine.: With warmup length $w = 100$ and floor fraction $\rho = 0.1000$:

$$\gamma_t = \begin{cases} \eta t / w & t \leq w \\ \eta \left[\rho + (1 - \rho) \frac{1}{2} \left(1 + \cos \pi \frac{t-w}{T-w} \right) \right] & t > w. \end{cases} \quad (3)$$

d) Step decay.: $\gamma_t = \eta \alpha^{k(t)}$ with $\alpha = 0.3000$ and $k(t)$ the number of drops passed, at steps 500 and 1000; no warmup.

e) Schedule-Free AdamW (reimplemented).: Following [9] with weighting exponent $r = 0$ and squared-learning-rate weights, we keep base iterate z_t and average x_t , take gradients at $y_t = (1 - \beta_1) z_t + \beta_1 x_t$, and update

$$z_{t+1} = z_t - \gamma_t \left(\frac{g_t}{\sqrt{v_t / (1 - \beta_2^t)} + \epsilon} + \lambda y_t \right), \quad (4)$$

$$c_t = \frac{\gamma_t^2}{\sum_{i \leq t} \gamma_i^2}, \quad x_{t+1} = (1 - c_t) x_t + c_t z_{t+1}, \quad (5)$$

where γ_t is a linear warmup to η over w steps and then constant. Weight decay is applied (as in the reference form) to matrices only. Evaluation uses x_T ; we also log the loss at y_T (val_loss_y). Because z and x move at different rates, the same peak LR has a different meaning here than for the other schedules. Note that the other schedules use β_1 momentum inside the Adam step, whereas Schedule-Free uses β_1 for the interpolation (no separate momentum buffer), as in the original.

f) Sensitivity metric.: For a system and seed, let $L(\eta)$ be the final validation loss. $\text{sens}_3 = \max_{\eta \in \mathcal{H}_3} L - \min_{\eta \in \mathcal{H}_3} L$ over the three main rates $\mathcal{H}_3 = \{10^{-3}, 3 \times 10^{-3}, 10^{-2}\}$ and sens_5 is $\max_{\eta \in \mathcal{H}_5} L - \min_{\eta \in \mathcal{H}_5} L$ over the five-rate grid (adding 3×10^{-4} and 3×10^{-2}). Both are computed per seed, using that seed's runs, and then averaged. The metric is a range on a coarse grid and therefore depends on where the grid sits relative to each method's optimum; Section VII discusses this.

IV. Experimental Setup

a) Data and model.: Tiny Shakespeare (1115394 characters, vocabulary 65); the first 90% of the text is used for training and the remaining 10% as validation. The model is a pre-LayerNorm GPT with 2 blocks, 4 heads, width 128, a $4 \times$ GELU MLP, learned positional embeddings, no dropout, and 421632 parameters. Training uses random windows of 64 characters, batch size 32, for 1500 steps so each run processes 3072000 characters, roughly three passes over the training text; the models are far from converged and no regularisation question arises (no dropout is used).

b) Metric.: Validation loss is the mean per-character cross-entropy (nats) on 256 fixed validation windows (identical for all runs), evaluated once after the last step (the averaged iterate for Schedule-Free). Training loss is the mean over the last 100 steps. A run is flagged diverged if the final validation loss exceeds 4 nats or is non-finite; no run in any group was flagged. There is no separate test split: no model selection other than the studied peak-LR axis is performed on the validation loss, and no hyperparameter was tuned (all non-LR settings were fixed before the runs; see Table I).

c) Grid and seeds.: Four systems $\times$ three main peak LRs ($10^{-3}, 3 \times 10^{-3}, 10^{-2}$) $\times$ 3 seeds form the core grid; the extra rates 3×10^{-4} and 3×10^{-2} ($\times$ four systems $\times$ 3 seeds) were run afterwards with the identical protocol; all five rates are logged in one registry group, main. The seed fixes initialisation and batch order. We used three rather than five seeds because each run takes roughly 90–120 seconds on the shared CPU and the whole study logged 112 runs; five seeds were not cheap enough. The ablations (Sec. VI) add warmup variants of cosine, constant and step decay at the same seeds. In total the study logged 112 runs (including derived sensitivity rows and four curve runs at seed 3).

TABLE I
Fixed hyperparameters (not tuned).

Setting	Value
Batch / context	32 / 64
Steps	1500
Adam β_1, β_2	0.9000, 0.9500
Weight decay (matrices)	0.1000
Grad-norm clip	1
Warmup (cosine, SF)	100 steps
Cosine floor	0.1000 $\times$ peak
Step decay	$\times 0.3000$ at 500, 1000

TABLE II

Main results: final validation loss (nats, mean $\pm$ std over 3 seeds) at the three main peak learning rates (10^{-3} , 3×10^{-3} , 10^{-2}) and the two extra rates (3×10^{-4} , 3×10^{-2}). Constant and step decay have no warmup here.

Method	Task	val_loss $\downarrow$	train_loss $\downarrow$	val_loss_y $\downarrow$	n
Step decay	lr3e-3	1.85 ± 0.05	1.70 ± 0.05	–	3
Schedule-free AdamW	lr3e-3	1.69 ± 0.01	1.53 ± 0.00	1.72 ± 0.01	3
Constant	lr3e-3	1.82 ± 0.03	1.66 ± 0.03	–	3
Warmup+Cosine	lr3e-3	1.67 ± 0.01	1.49 ± 0.00	–	3
Step decay	lr1e-3	1.83 ± 0.00	1.68 ± 0.01	–	3
Schedule-free AdamW	lr1e-3	1.88 ± 0.01	1.75 ± 0.01	1.88 ± 0.01	3
Constant	lr1e-3	1.76 ± 0.01	1.60 ± 0.00	–	3
Warmup+Cosine	lr1e-3	1.77 ± 0.01	1.61 ± 0.01	–	3
Step decay	lr1e-2	2.35 ± 0.06	2.32 ± 0.05	–	3
Schedule-free AdamW	lr1e-2	1.63 ± 0.00	1.49 ± 0.00	1.67 ± 0.01	3
Constant	lr1e-2	2.42 ± 0.05	2.40 ± 0.04	–	3
Warmup+Cosine	lr1e-2	1.68 ± 0.01	1.50 ± 0.00	–	3
Step decay	lr3e-4	2.13 ± 0.01	2.09 ± 0.01	–	3
Schedule-free AdamW	lr3e-4	2.17 ± 0.01	2.13 ± 0.01	2.17 ± 0.01	3
Constant	lr3e-4	1.94 ± 0.02	1.83 ± 0.01	–	3
Warmup+Cosine	lr3e-4	2.07 ± 0.01	2.02 ± 0.01	–	3
Step decay	lr3e-2	2.50 ± 0.00	2.48 ± 0.00	–	3
Schedule-free AdamW	lr3e-2	1.65 ± 0.02	1.55 ± 0.00	1.71 ± 0.01	3
Constant	lr3e-2	2.80 ± 0.05	2.76 ± 0.03	–	3
Warmup+Cosine	lr3e-2	1.71 ± 0.01	1.54 ± 0.00	–	3

d) Hardware and fairness.: All runs use two CPU threads on a shared Apple-silicon machine, at most two processes at a time. Every system goes through the same script (method/run.py), the same data order function, evaluation set and clipping; the only differences are those in Section III. We did not tune warmup length, decay shape, step positions or the Schedule-Free parameters; the peak-LR grid is the only axis explored, so every ranking is relative to our fixed choices.

V. Results

a) H1 (cosine beats constant at the best LR): supported, but conditional.: Taking each system’s best of the three main rates, warmup+cosine reaches 1.67 at 3×10^{-3} , constant LR 1.76 at 10^{-3} , step decay 1.83 at 10^{-3} and Schedule-Free 1.63 at 10^{-2} (Table II). The comparison of per-seed best losses, warmup+cosine minus constant, is -0.092 nats (Welch $p = 0.001443$, $n = 3$ per arm; this follows the test registered in research.yaml). The advantage is not uniform across peak rates: at 10^{-3} the two are indistinguishable ($\Delta = 0.001$, $p = 0.9139$), and at the lowest rate 3×10^{-4} constant LR is better than warmup+cosine (1.94 vs. 2.07; $p = 0.0004348$): when the

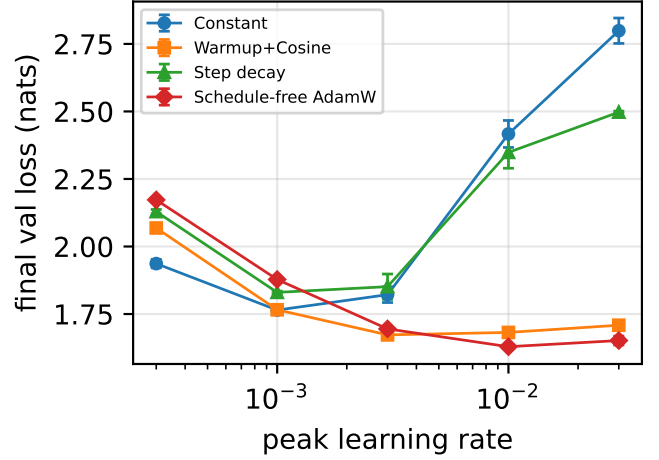


Fig. 1. Final validation loss against peak learning rate (log axis), mean and std over 3 seeds, five rates (group main).

TABLE III

Sensitivity: worst-minus-best validation loss over three (sens₃) and five (sens₅) peak rates, and best three-rate loss (best₃), mean $\pm$ std over seeds (lower is better).

Method	Task	sens3 $\downarrow$	sens5 $\downarrow$	best3 $\downarrow$	n
Step decay	sens	0.53 ± 0.05	0.68 ± 0.02	1.82 ± 0.02	3
Schedule-free AdamW	sens	0.25 ± 0.01	0.54 ± 0.01	1.63 ± 0.00	3
Constant	sens	0.65 ± 0.05	1.03 ± 0.03	1.76 ± 0.01	3
Warmup+Cosine	sens	0.09 ± 0.01	0.40 ± 0.02	1.67 ± 0.01	3

peak rate is too small for a 1,500-step budget, decaying it only removes useful progress. At 3×10^{-3} and 10^{-2} cosine wins by -0.15 and -0.73 nats respectively, but the 10^{-2} gap is largely a warmup effect (below).

b) Schedule-Free beats cosine at its best rate, but not everywhere.: At 10^{-2} Schedule-Free AdamW is lower than warmup+cosine (cosine minus Schedule-Free: 0.053 nats; Welch $p = 0.002789$) and likewise at 3×10^{-2} (1.65 vs. 1.71). At 3×10^{-3} the two are close (1.69 vs. 1.67, Welch $p = 0.04281$), and at lower rates Schedule-Free is clearly worse: 1.88 at 10^{-3} and 2.17 at 3×10^{-4} , below even the constant schedule at those rates. Its loss at the non-averaged iterate y_T is worse than at the average at the best rates (e.g. 1.67 at 10^{-2}), consistent with the averaging doing useful work.

c) H2 (Schedule-Free less LR-sensitive than cosine): refuted.: Table III gives sens₃ of 0.25 for Schedule-Free and 0.09 for warmup+cosine (difference cosine minus Schedule-Free -0.16 , Welch $p = 0.0008339$); over five rates the values are 0.54 and 0.40 ($p = 0.0006695$). The direction is opposite to H2. Figure 1 shows why: Schedule-Free is flat and good for 10^{-2} and 3×10^{-2} but degrades quickly below 3×10^{-3} , whereas cosine has a broad, shallow basin around 3×10^{-3} . Warmup+cosine is the least sensitive of the four as specified (0.65 for constant, 0.53 for step

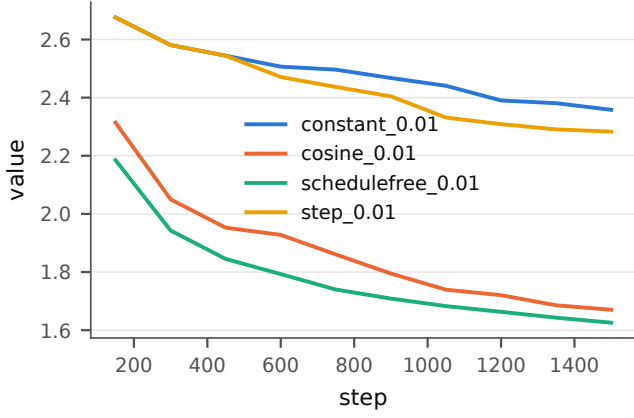


Fig. 2. Validation loss during training at peak LR 10^{-2} (seed 3, one run per system; evaluated every 150 steps; Schedule-Free evaluated at the averaged iterate).

decay over three rates), but this ordering changes once the baselines get warmup (next section). Note that sensitivity is a range on a coarse grid: Schedule-Free’s best loss is at or beyond the upper end of the main grid, so a grid shifted upward would make its sens_3 smaller; our result is about this grid, not about an intrinsic robustness property.

d) Training dynamics at 10^{-2} .: Figure 2 shows that without warmup, constant and step decay (identical until the first drop) stay far above the schedules with warmup throughout training (final validation loss 2.36 for constant and 2.28 for step decay, against 1.67 for cosine and 1.63 for Schedule-Free in these runs). This is single-seed evidence for the early-training failure; it motivates, but does not by itself establish, the warmup explanation tested in the next section.

e) Failure cases and per-regime summary.: (i) Constant and step decay at 10^{-2} and above: 2.42 and 2.35 nats, 2.80 and 2.50 at 3×10^{-2} ; no run was flagged as diverged, they simply train badly. (ii) Warmup+cosine at 3×10^{-4} (2.07), where it is worse than constant. (iii) Schedule-Free at $\leq 10^{-3}$. No system is best at all rates; the ranking depends on the regime, which is the main message of the sweep table.

VI. Ablations and Analysis

a) Warmup length at peak LR 10^{-2} (H3).: Table IV varies the warmup length. Cosine without warmup reaches 2.25 nats, against 1.69 with 50 warmup steps, 1.68 with the default 100 (main group) and 1.67 with 300. Relative to the no-warmup run, the difference with 50 steps is 0.55 nats (Welch $p = 0.001071$) and with 300 steps 0.57 ($p = 0.001287$). H3 is supported. The registered comparison was against the 100-step default, which lives in the main group and which the comparison tool cannot pool with the ablation group; we therefore tested against the 50- and 300-step runs that flank it and report the 100-step mean from the main table. Beyond roughly 50 steps, more

TABLE IV
Warmup ablation at peak LR 10^{-2} (validation loss).

Method	val_loss ↓	n
Warmup+Cosine (warmup=50)	1.69 ± 0.01	3
Warmup+Cosine (warmup=0)	2.25 ± 0.04	3
Constant (warmup=100)	1.81 ± 0.00	3
Constant (warmup=300)	1.80 ± 0.00	3
Warmup+Cosine (warmup=300)	1.67 ± 0.00	3

TABLE V
Step decay and constant LR with a 100-step warmup at the three main rates (constant at 10^{-2} is in Table IV).

Method	Task	val_loss ↓	n
Step decay (warmup=100)	lr3e-3	1.71 ± 0.00	3
Constant (warmup=100)	lr3e-3	1.74 ± 0.01	3
Step decay (warmup=100)	lr1e-3	1.83 ± 0.01	3
Constant (warmup=100)	lr1e-3	1.76 ± 0.02	3
Step decay (warmup=100)	lr1e-2	1.71 ± 0.01	3

warmup changes little at this rate (the means for 50, 100 and 300 steps are listed above and lie within about two seed standard deviations of each other; we did not test these differences).

b) Is it the schedule or the warmup?: The same table shows that giving constant LR a warmup of 100 or 300 steps lowers its loss at 10^{-2} from 2.42 to 1.81 and 1.80, so most of the gap between cosine and constant at this rate is attributable to warmup rather than decay; the remaining gap (cosine 1.68 vs. constant with warmup 100 at 1.81) is the decay’s contribution. This is a controlled isolation of the warmup factor (same code, one flag), so we state it as a result and not only as a conjecture; why warmup helps (e.g. Adam’s early second-moment estimate [6]) is not tested here.

c) Fair-warmup baselines and sensitivity.: Table V adds a 100-step warmup to step decay at all three main rates and to constant LR at 10^{-3} and 3×10^{-3} (constant at 10^{-2} is in the warmup table). With warmup, step decay’s loss at its best rate is 1.70 (versus 1.82 without) and constant’s is 1.74 (versus 1.76). Their sens_3 drops to 0.13 and 0.06, so constant LR with warmup has the lowest three-rate sensitivity of any configuration we ran, slightly below warmup+cosine’s 0.09 (descriptive; no test was run on this comparison, and the ordering could change on a different grid). The cost is a higher best loss than cosine (1.67) and Schedule-Free (1.63). Hence the sensitivity advantage reported for cosine in the previous section belongs largely to warmup, and the schedule shape (decay) mostly buys a lower loss at a given rate. We did not run these fair-warmup variants at 3×10^{-4} or 3×10^{-2} , so sens_5 is only available for the four main systems.

d) Sensitivity to the peak LR (sweep).: The five-rate sweep (Fig. 1, Table II) is itself the second sensitivity study: each system’s loss rises on both sides of its best grid point, but the largest rate 3×10^{-2} is still good for warmup+cosine (1.71) and Schedule-Free (1.65), so the optimum of those two lies near or above the upper end of the grid, and a higher rate than 3×10^{-2} was not tried.

VII. Limitations

- Scale. One model (421632 parameters), one dataset, one budget of 1500 steps (far from convergence). Rankings between schedules are known to change with training length, and a short budget favours schedules that reach low loss quickly.
- Seeds and statistics. Only 3 seeds per cell, so Welch/paired p -values rest on $n = 3$ and the standard deviations are themselves noisy; many comparisons are decided by effect sizes that are large relative to the seed spread, but small differences (such as between cosine at neighbouring peak rates) are not resolved. We ran many comparisons without multiplicity correction.
- Untuned secondary choices. Warmup length, cosine floor, step positions and factor, Adam betas, weight decay and the Schedule-Free configuration were fixed a priori, not tuned. A different Schedule-Free warmup or weighting could change its sensitivity; we cannot say whether its poor low-LR behaviour is intrinsic or due to our choices (the interplay of the squared-LR weights and the 100-step warmup may matter).
- Reimplementation. All four systems, including Schedule-Free AdamW, are written in method/run.py from the published description; they were not checked against the authors’ reference implementation numerically.
- Confounded baseline specification. As the seed asked, constant and step decay have no warmup in the main table. We added a warmup ablation to separate the effects, but the fair-warmup runs cover only some cells (both only at the three main rates), and sensitivity of those variants is over three rates.
- Sensitivity metric. A range over a coarse five-point log grid whose position relative to each optimum matters; Schedule-Free’s best rate is at or above the upper end of the main grid. We did not measure sensitivity to other hyperparameters or interactions of peak LR with the decay horizon.
- Single metric. We report validation loss on 256 fixed windows after the last step only; we did not evaluate at equal wall-clock (Schedule-Free’s extra state costs memory and some time), generalisation gaps, or sample quality, and no test set was held out because no model selection was done on it. Learning-rate curves in Fig. 2 are single-seed.
- Mechanism. We isolated warmup as a factor; we did not test why it helps.

VIII. Conclusion

In a two-layer character-level GPT trained for 1500 steps, warmup+cosine beats constant LR at its best peak rate, but the advantage vanishes at a low rate (10^{-3}) and reverses at 3×10^{-4} , and a large part of the apparent advantage at high peak rates is explained by warmup, which the constant and step-decay baselines lacked as specified. Schedule-Free AdamW (our reimplementation) reached the lowest loss (1.63) but was more sensitive to the peak LR over our grid than warmup+cosine, so the hypothesis that it is easier to tune is refuted here. Constant LR with a modest warmup had the smallest three-rate sensitivity of all configurations, at a cost in loss. These findings are for one tiny model, three seeds, and untuned secondary hyperparameters; a larger study would add more seeds, tune warmup per schedule, extend the peak-LR grid upward, test other budgets, and compare the reimplementation with the reference Schedule-Free code.

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